```python
import numpy as np
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt
from sklearn.preprocessing import LabelEncoder, OneHotEncoder

df = pd.read_csv('insurance.csv')
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

EDA

```python
df.shape
```

```
(1338, 7)
```

```python
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```python
df.describe()
```

```
               age          bmi     children       charges
count  1338.000000  1338.000000  1338.000000   1338.000000
mean     39.207025    30.663397     1.094918  13270.422265
std      14.049960     6.098187     1.205493  12110.011237
min      18.000000    15.960000     0.000000   1121.873900
25%      27.000000    26.296250     0.000000   4740.287150
50%      39.000000    30.400000     1.000000   9382.033000
75%      51.000000    34.693750     2.000000  16639.912515
max      64.000000    53.130000     5.000000  63770.428010
```

```
df.isnull().sum()
```

```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

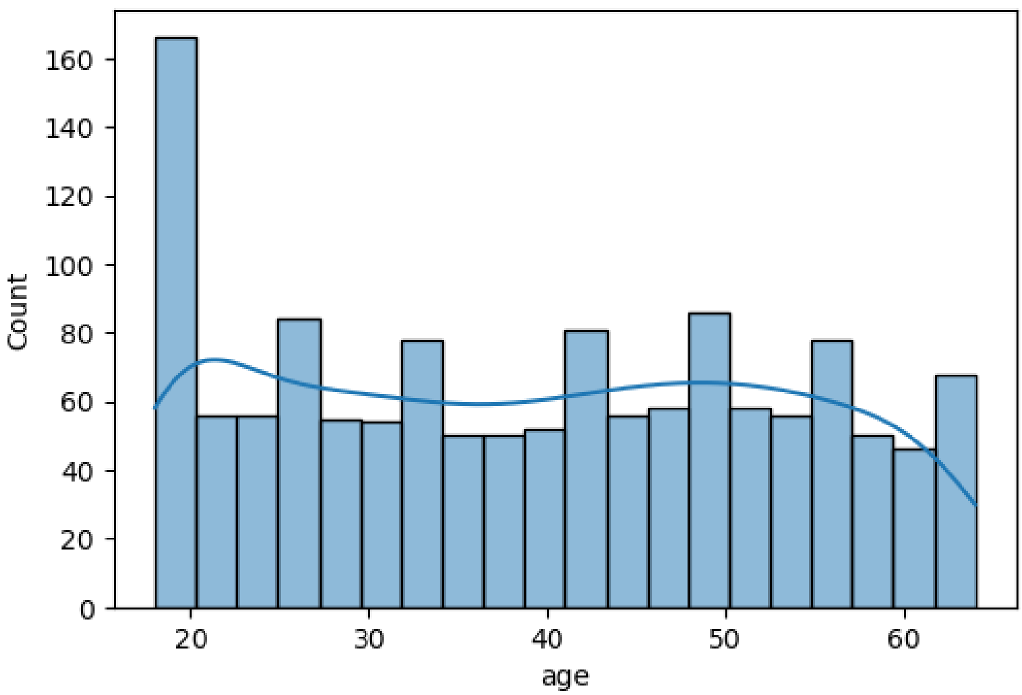
```
df.columns
```

```
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region',
'charges'], dtype='object')
```

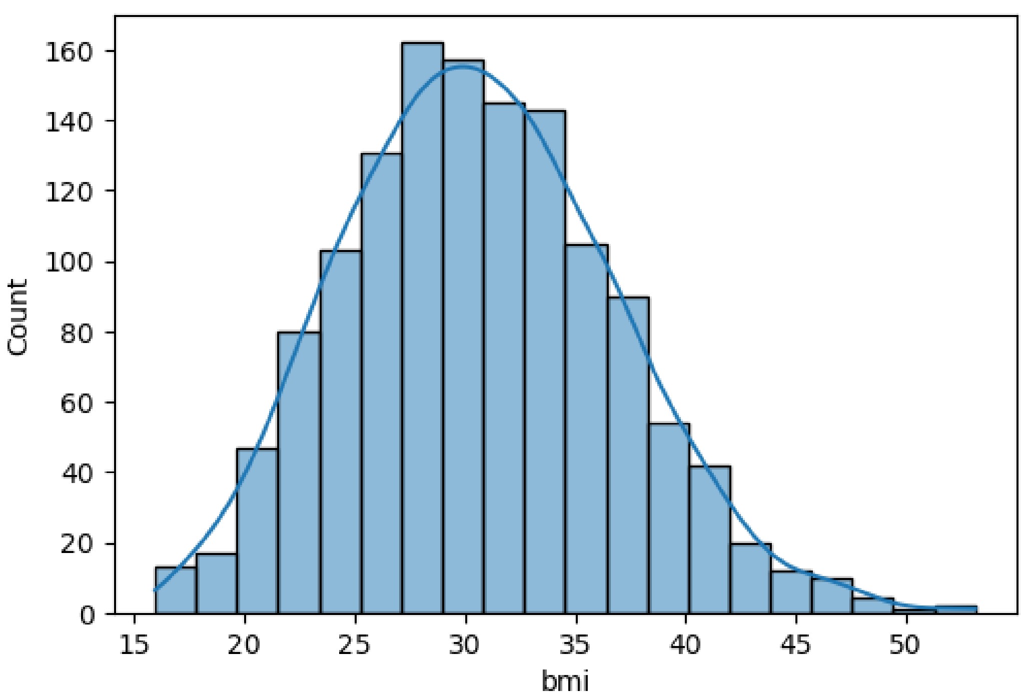
```
numericals = df.select_dtypes(include = ['int',
'float']).columns.tolist()
numericals
```

```
['age', 'bmi', 'children', 'charges']
```

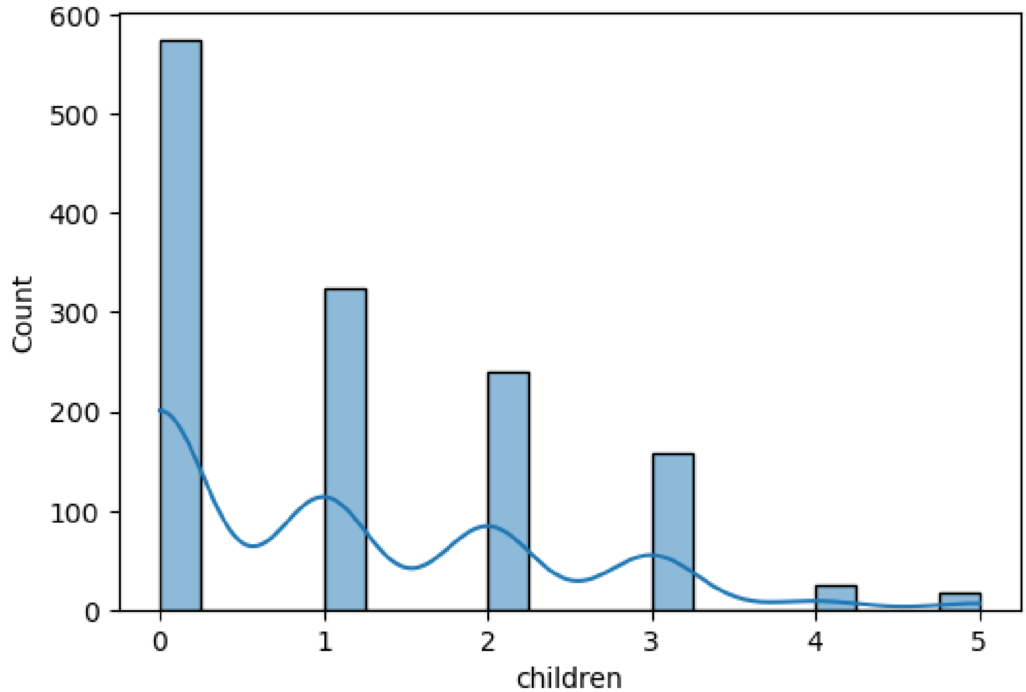
```
for i in numericals:
    plt.figure(figsize=(6,4))
    sns.histplot(df[i], kde = True, bins = 20)
```

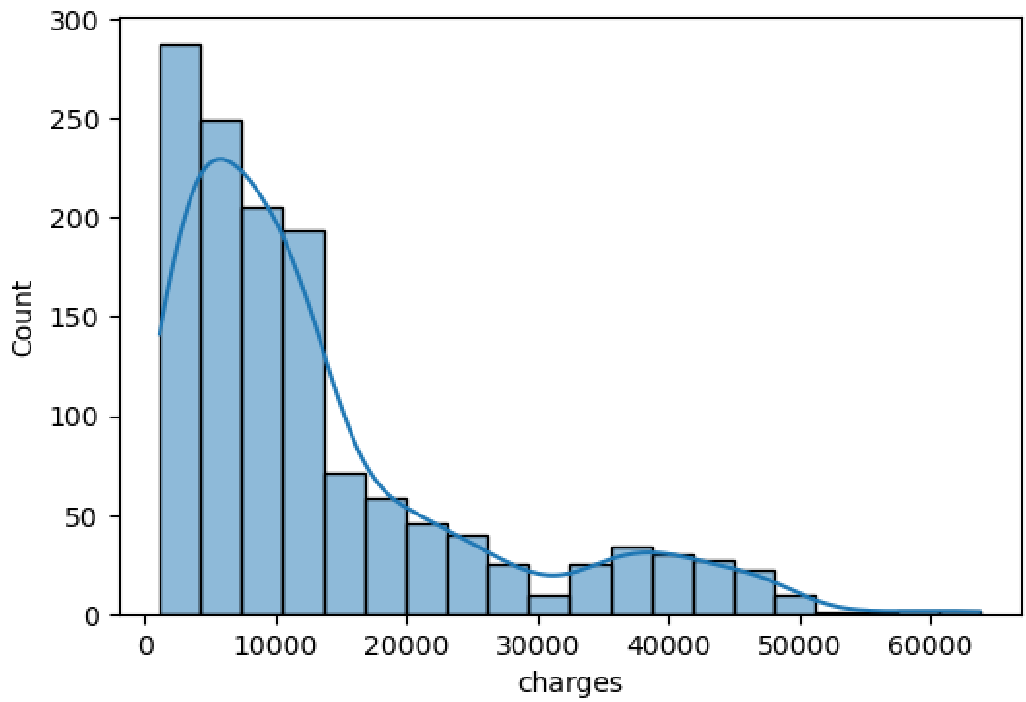


Count
age
0
20
40
60
80
100
120
140
160
20
30
40
50
60



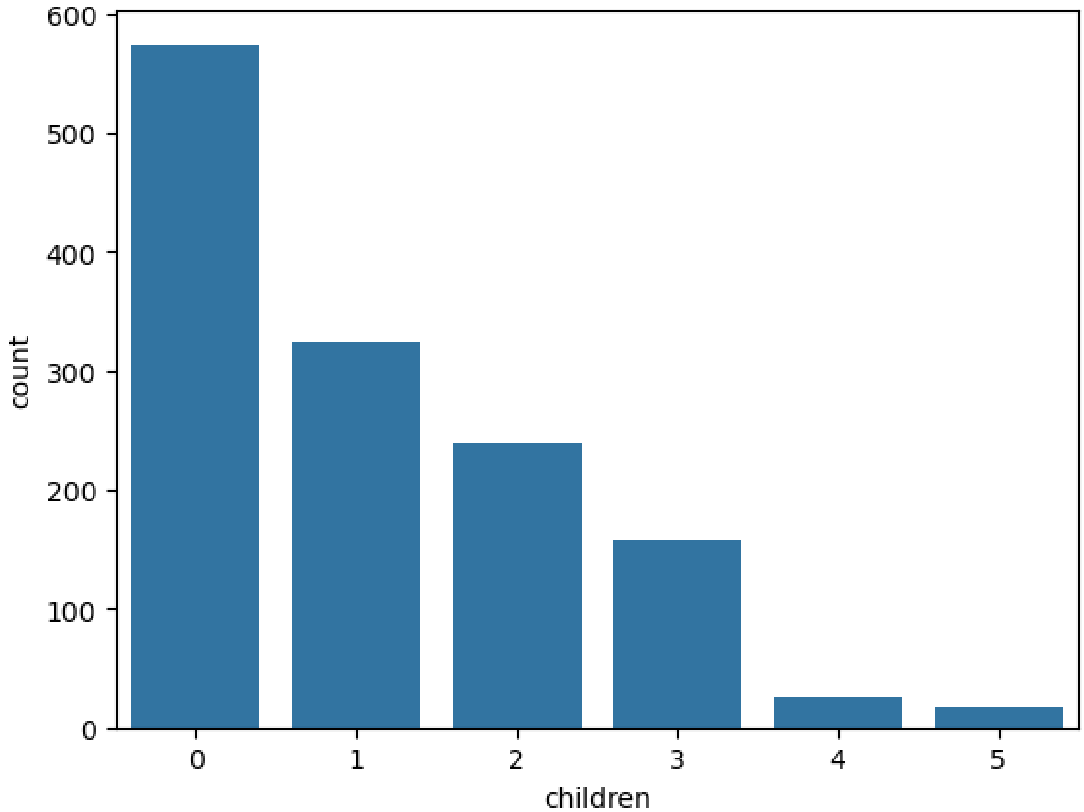
Count
bmi
0
20
40
60
80
100
120
140
160
15
20
25
30
35
40
45
50





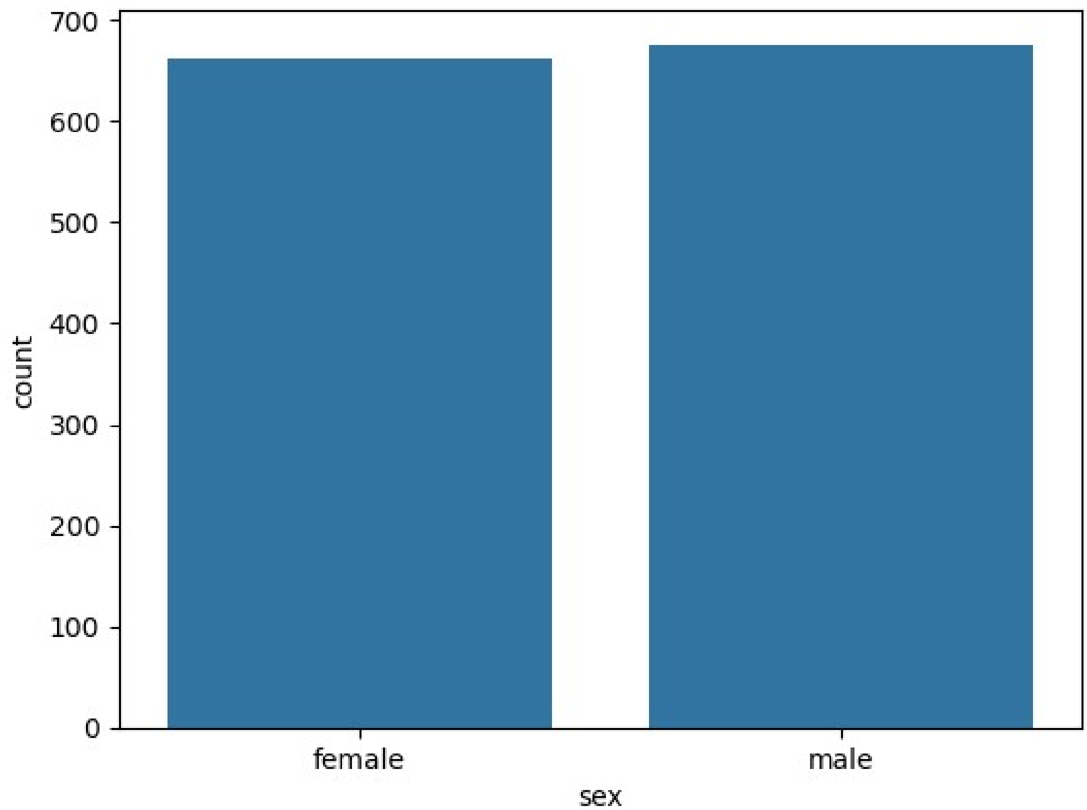

```
sns.countplot(x = df['children'])
```

```
<Axes: xlabel='children', ylabel='count'>
```



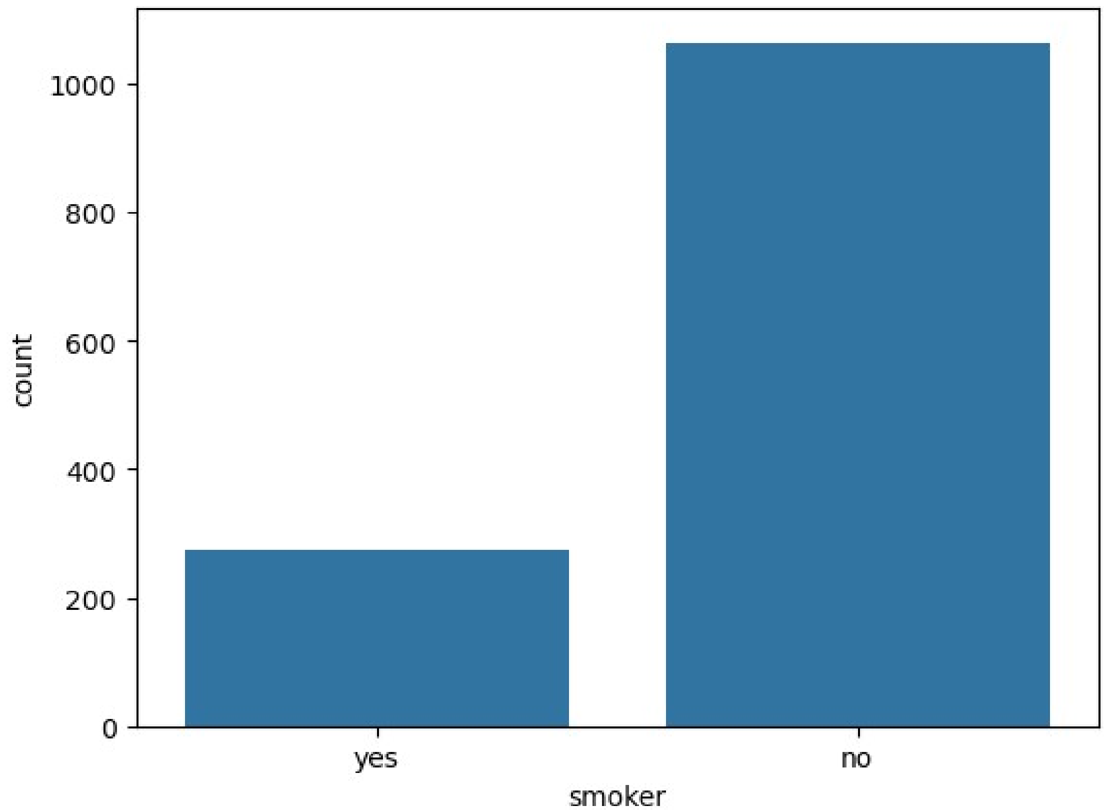

```
sns.countplot(x = df['sex'])
```

```
<Axes: xlabel='sex', ylabel='count'>
```

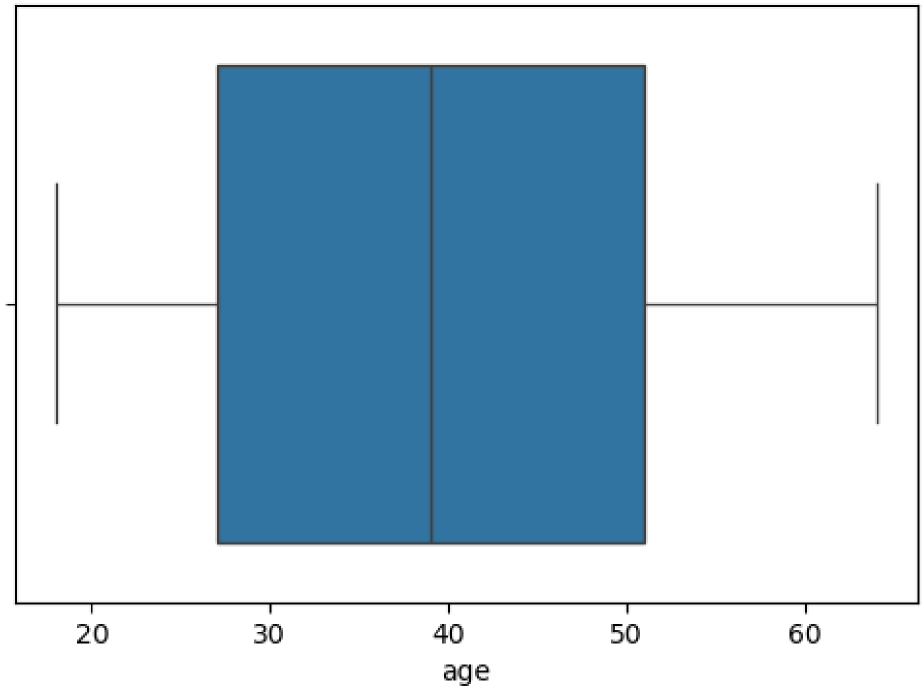


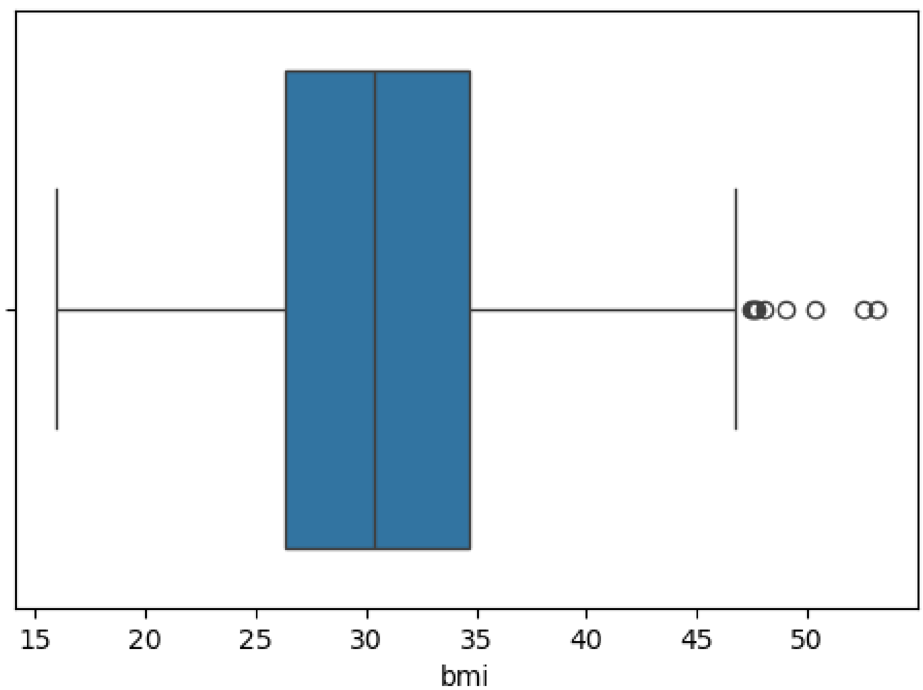

```
sns.countplot(x = df['smoker'])
```

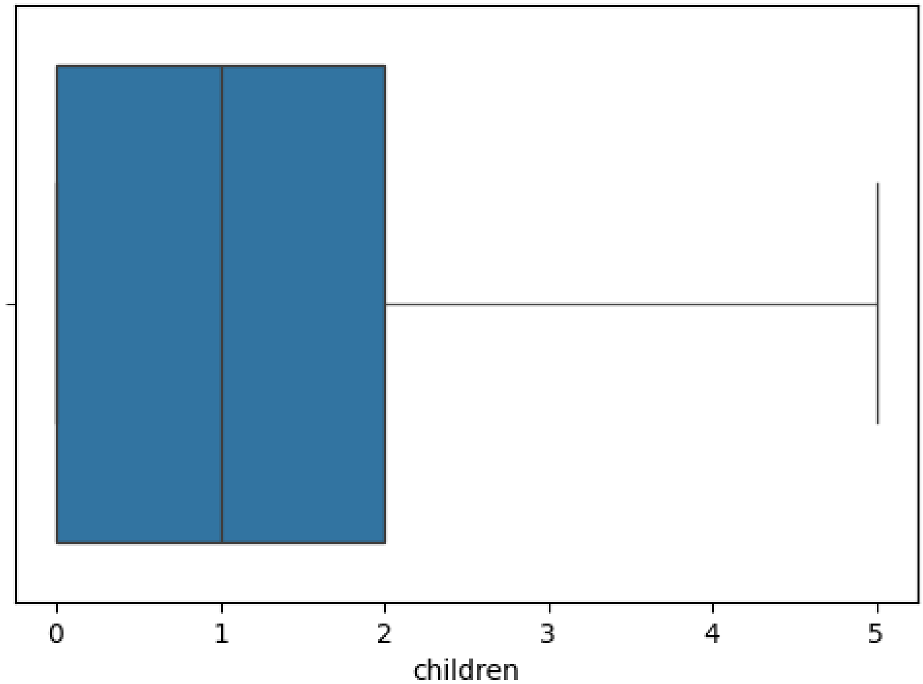
```
<Axes: xlabel='smoker', ylabel='count'>
```

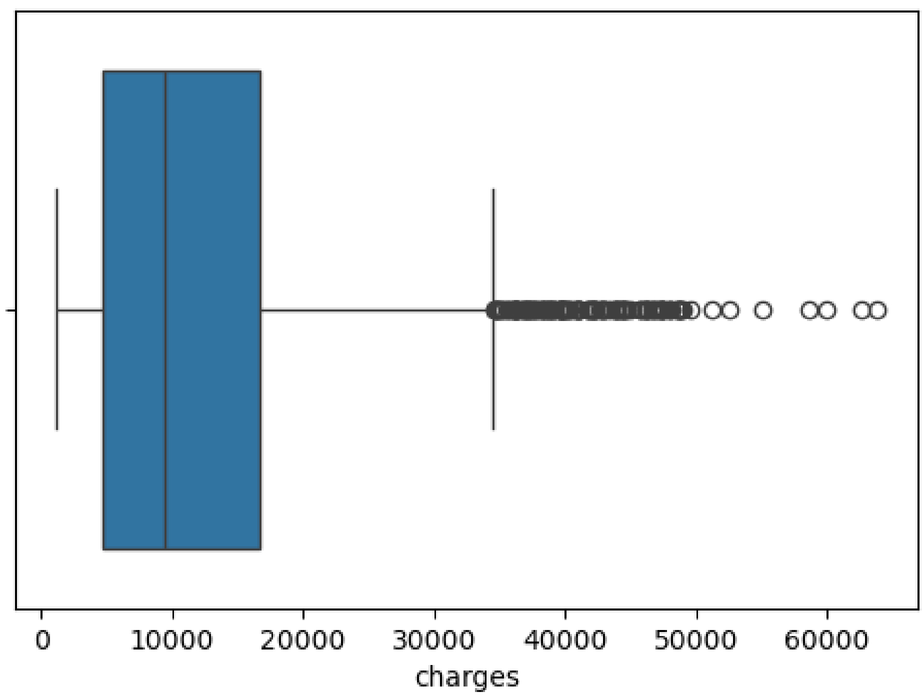



```
for col in numericals:
    plt.figure(figsize= (6,4))
    sns.boxplot(x = df[col])
```


20
30
40
50
60
age

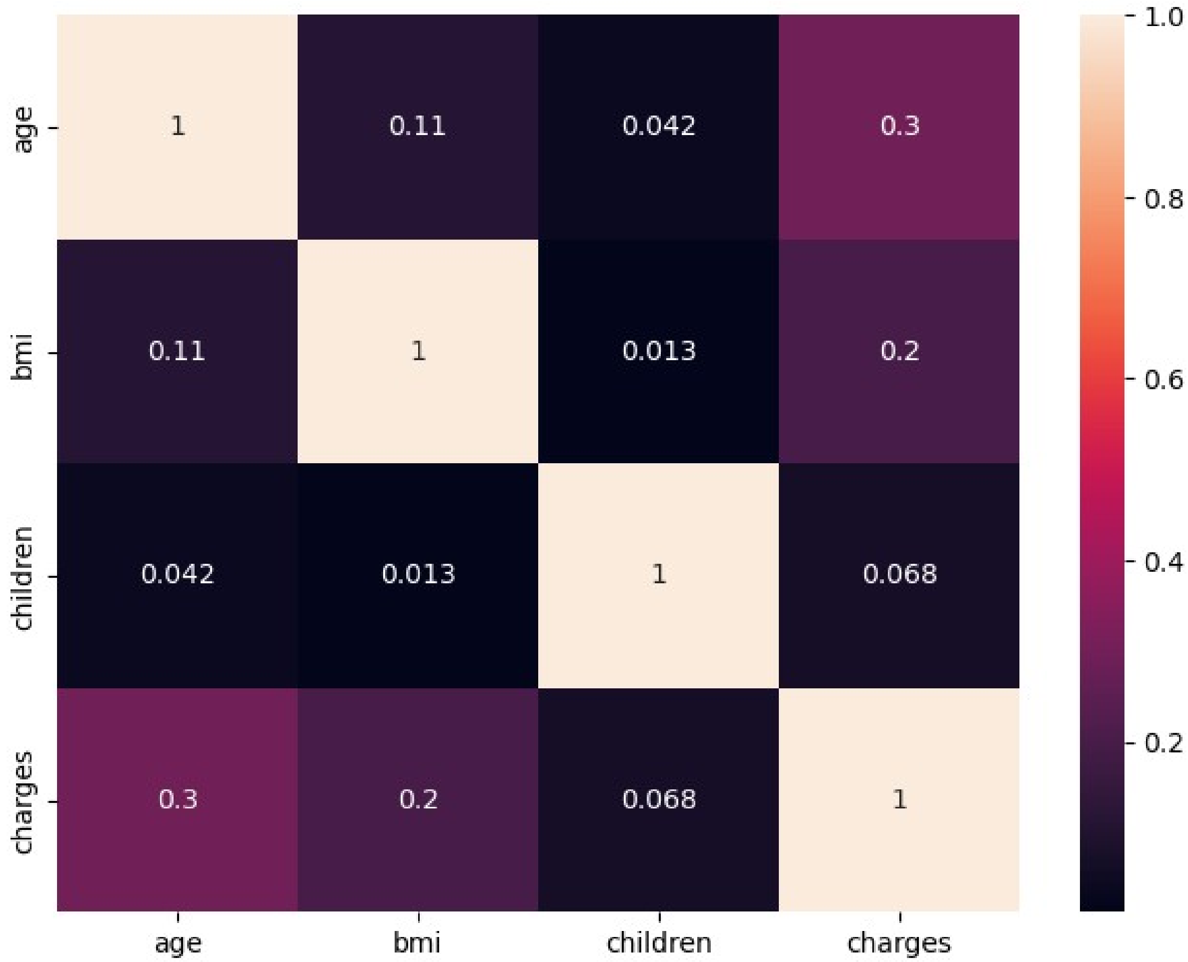

15
20
25
30
35
40
45
50
bmi






```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True),annot=True)
```

```
<Axes: >
```



Data Cleaning & Preprocessing

```
df_cleaned = df.copy()
```

```
df_cleaned.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
df_cleaned.shape
```

```
(1338, 7)
```

```
df_cleaned.drop_duplicates(inplace = True)
df_cleaned.shape
```

```
(1337, 7)
```

```python
df_cleaned.isnull().sum()
```

```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

```python
df_cleaned.dtypes
```

```
age           int64
sex          object
bmi         float64
children      int64
smoker       object
region       object
charges     float64
dtype: object
```

```python
df_cleaned['sex'].value_counts()
```

```
sex
male      675
female    662
Name: count, dtype: int64
```

```python
df_cleaned['sex'] = df_cleaned['sex'].map({"male" : 0,"female" : 1})
```

```python
df_cleaned.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	1	27.900	0	yes	southwest	16884.92400
1	18	0	33.770	1	no	southeast	1725.55230
2	28	0	33.000	3	no	southeast	4449.46200
3	33	0	22.705	0	no	northwest	21984.47061
4	32	0	28.880	0	no	northwest	3866.85520

```python
df_cleaned['smoker'].value_counts()
```

```
smoker
no     1063
yes     274
Name: count, dtype: int64
```

```python
df_cleaned['smoker'] = df_cleaned['smoker'].map({"no" : 0,"yes" : 1})
```

```python
df_cleaned
```

```
      age  sex     bmi  children  smoker     region      charges
0      19    1  27.900         0       1  southwest  16884.92400
1      18    0  33.770         1       0  southeast   1725.55230
2      28    0  33.000         3       0  southeast   4449.46200
3      33    0  22.705         0       0  northwest  21984.47061
4      32    0  28.880         0       0  northwest   3866.85520
...   ...  ...     ...       ...     ...        ...          ...
1333   50    0  30.970         3       0  northwest  10600.54830
1334   18    1  31.920         0       0  northeast   2205.98080
1335   18    1  36.850         0       0  southeast   1629.83350
1336   21    1  25.800         0       0  southwest   2007.94500
1337   61    1  29.070         0       1  northwest  29141.36030

[1337 rows x 7 columns]
```

```
df_cleaned.rename(columns={
    'sex' :'is_female',
    'smoker': 'is_smoker'
                          },inplace = True)
```

```
df_cleaned.head()
```

```
   age  is_female     bmi  children  is_smoker     region      charges
0   19          1  27.900         0          1  southwest  16884.92400
1   18          0  33.770         1          0  southeast   1725.55230
2   28          0  33.000         3          0  southeast   4449.46200
3   33          0  22.705         0          0  northwest  21984.47061
4   32          0  28.880         0          0  northwest   3866.85520
```

```
df['region'].value_counts()
```

```
region
southeast    364
southwest    325
northwest    325
northeast    324
Name: count, dtype: int64
```

```
df_cleaned = pd.get_dummies(df_cleaned,columns =
['region'],drop_first=True)
```

```
df_cleaned.head()
```

```
   age  is_female     bmi  children  is_smoker      charges
region_northwest  \
0   19          1  27.900         0          1  16884.92400
False
1   18          0  33.770         1          0   1725.55230
False
2   28          0  33.000         3          0   4449.46200
False
```

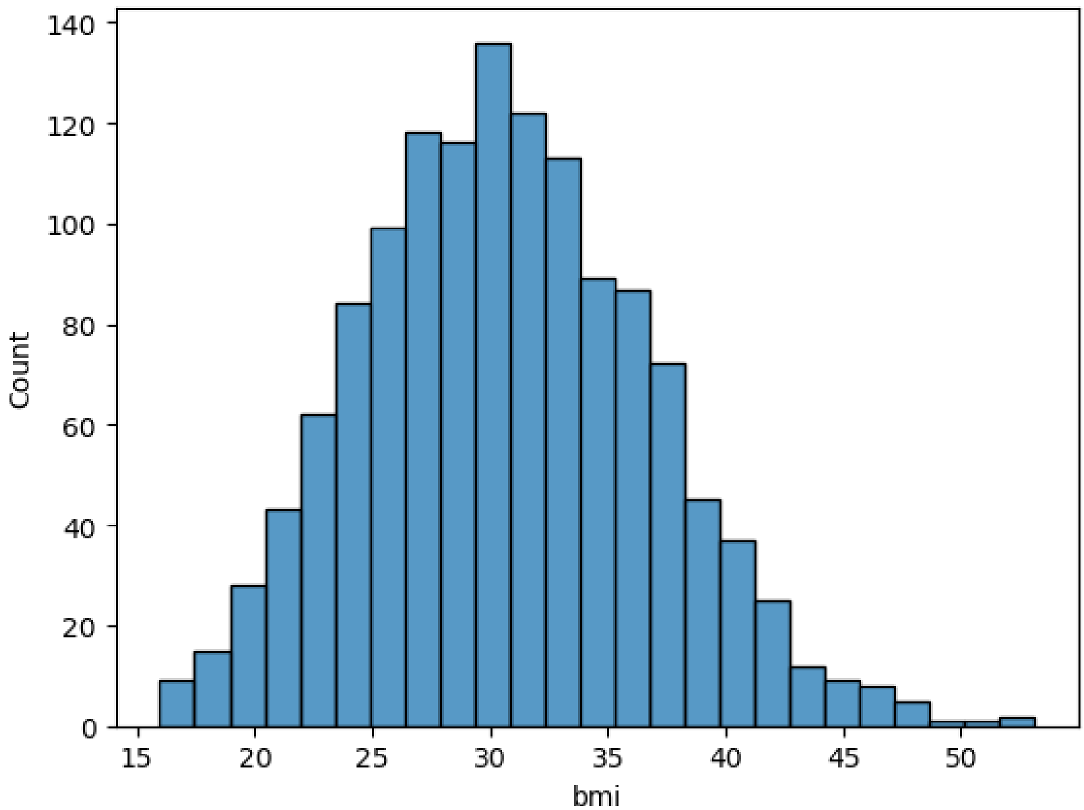
```
3   33            0  22.705         0           0  21984.47061
True
4   32            0  28.880         0           0   3866.85520
True

   region_southeast  region_southwest
0             False              True
1              True             False
2              True             False
3             False             False
4             False             False
```

```
df_cleaned = df_cleaned.astype(int)

df_cleaned.head(3)
```

```
   age  is_female  bmi  children  is_smoker  charges  region_northwest
\
0   19          1   27         0          1    16884                 0

1   18          0   33         1          0     1725                 0

2   28          0   33         3          0     4449                 0


   region_southeast  region_southwest
0                 0                 1
1                 1                 0
2                 1                 0
```

Feature Engineering and Extraction

```
sns.histplot(df['bmi'])

<Axes: xlabel='bmi', ylabel='Count'>
```




```
df_cleaned['bmi_category'] = pd.cut(
    df_cleaned['bmi'],
    bins=[0, 18.5, 24.9, 29.9, float('inf')],
    labels=['Underweight', 'Normal', 'Overweight', 'Obese']
)

df_cleaned.head(3)

   age  is_female  bmi  children  is_smoker  charges  region_northwest
\
0   19          1   27         0          1    16884                 0

1   18          0   33         1          0     1725                 0

2   28          0   33         3          0     4449                 0


   region_southeast  region_southwest bmi_category
0                 0                 1   Overweight
1                 1                 0        Obese
2                 1                 0        Obese

df_cleaned = pd.get_dummies(df_cleaned,columns =
['bmi_category'],drop_first=True)
```

```
df_cleaned = df_cleaned.astype(int)

df_cleaned.head()

   age  is_female  bmi  children  is_smoker  charges  region_northwest
\
0   19          1   27         0          1    16884                 0

1   18          0   33         1          0     1725                 0

2   28          0   33         3          0     4449                 0

3   33          0   22         0          0    21984                 1

4   32          0   28         0          0     3866                 1


   region_southeast  region_southwest  bmi_category_Normal  \
0                 0                 1                    0
1                 1                 0                    0
2                 1                 0                    0
3                 0                 0                    1
4                 0                 0                    0

   bmi_category_Overweight  bmi_category_Obese
0                        1                   0
1                        0                   1
2                        0                   1
3                        0                   0
4                        1                   0

df_cleaned.columns

Index(['age', 'is_female', 'bmi', 'children', 'is_smoker', 'charges',
       'region_northwest', 'region_southeast', 'region_southwest',
       'bmi_category_Normal', 'bmi_category_Overweight',
'bmi_category_Obese'],
      dtype='object')

from sklearn.preprocessing import StandardScaler
cols = ['age','bmi','children']
scaler = StandardScaler()

df_cleaned[cols] = scaler.fit_transform(df_cleaned[cols])

df_cleaned.head()

        age  is_female       bmi  children  is_smoker  charges  \
0 -1.440418          1 -0.517949 -0.909234          1    16884
1 -1.511647          0  0.462463 -0.079442          0     1725
2 -0.799350          0  0.462463  1.580143          0     4449
3 -0.443201          0 -1.334960 -0.909234          0    21984
```

```
4 -0.514431         0 -0.354547 -0.909234         0     3866

   region_northwest  region_southeast  region_southwest
bmi_category_Normal  \
0                 0                 0                 1
0
1                 0                 1                 0
0
2                 0                 1                 0
0
3                 1                 0                 0
1
4                 1                 0                 0
0

   bmi_category_Overweight  bmi_category_Obese
0                        1                   0
1                        0                   1
2                        0                   1
3                        0                   0
4                        1                   0
```

```
from scipy.stats import pearsonr

# ----------------------------------
# Pearson Correlation Calculation
# ----------------------------------

# List of features to check against target
selected_features = [
    'age', 'bmi', 'children', 'is_female', 'is_smoker',
    'region_northwest', 'region_southeast', 'region_southwest',
    'bmi_category_Normal', 'bmi_category_Overweight',
'bmi_category_Obese'
]

correlations = {
    feature: pearsonr(df_cleaned[feature], df_cleaned['charges'])[0]
    for feature in selected_features
}
correlation_df = pd.DataFrame(list(correlations.items()),
columns=['Feature', 'Pearson Correlation'])
correlation_df.sort_values(by='Pearson Correlation', ascending=False)
```

```
                   Feature  Pearson Correlation
4                is_smoker             0.787234
0                      age             0.298309
10      bmi_category_Obese             0.200348
1                      bmi             0.196236
6         region_southeast             0.073577
```

2	children	0.067390
5	region_northwest	-0.038695
7	region_southwest	-0.043637
3	is_female	-0.058046
8	bmi_category_Normal	-0.104042
9	bmi_category_Overweight	-0.120601

```python
cat_features = [
    'is_female', 'is_smoker',
    'region_northwest', 'region_southeast', 'region_southwest',
    'bmi_category_Normal', 'bmi_category_Overweight',
'bmi_category_Obese'
]

from scipy.stats import chi2_contingency
import pandas as pd

alpha = 0.05

df_cleaned['charges_bin'] = pd.qcut(df_cleaned['charges'], q=4,
labels=False)
chi2_results = {}

for col in cat_features:
    contingency = pd.crosstab(df_cleaned[col],
df_cleaned['charges_bin'])
    chi2_stat, p_val, _, _ = chi2_contingency(contingency)
    decision = 'Reject Null (Keep Feature)' if p_val < alpha else
'Accept Null (Drop Feature)'
    chi2_results[col] = {
        'chi2_statistic': chi2_stat,
        'p_value': p_val,
        'Decision': decision
    }

chi2_df = pd.DataFrame(chi2_results).T
chi2_df = chi2_df.sort_values(by='p_value')
chi2_df
```

	chi2_statistic	p_value	Decision
is_smoker	848.219178	0.0	Reject Null (Keep Feature)
region_southeast	15.998167	0.001135	Reject Null (Keep Feature)
is_female	10.258784	0.01649	Reject Null (Keep Feature)
bmi_category_Obese	8.515711	0.036473	Reject Null (Keep Feature)
region_southwest	5.091893	0.165191	Accept Null (Drop

```
Feature)
bmi_category_Overweight        4.25149  0.235557  Accept Null (Drop
Feature)
bmi_category_Normal           3.708088   0.29476  Accept Null (Drop
Feature)
region_northwest               1.13424  0.768815  Accept Null (Drop
Feature)
```

```
final_df = df_cleaned[['age', 'is_female', 'bmi', 'children',
'is_smoker', 'charges','region_southeast','bmi_category_Obese']]
final_df
```

```
           age  is_female       bmi  children  is_smoker  charges  \
0    -1.440418          1 -0.517949 -0.909234          1    16884
1    -1.511647          0  0.462463 -0.079442          0     1725
2    -0.799350          0  0.462463  1.580143          0     4449
3    -0.443201          0 -1.334960 -0.909234          0    21984
4    -0.514431          0 -0.354547 -0.909234          0     3866
...        ...        ...       ...       ...        ...      ...
1333  0.767704          0 -0.027743  1.580143          0    10600
1334 -1.511647          1  0.135659 -0.909234          0     2205
1335 -1.511647          1  0.952670 -0.909234          0     1629
1336 -1.297958          1 -0.844753 -0.909234          0     2007
1337  1.551231          1 -0.191145 -0.909234          1    29141

      region_southeast  bmi_category_Obese
0                    0                   0
1                    1                   1
2                    1                   1
3                    0                   0
4                    0                   0
...                ...                 ...
1333                 0                   1
1334                 0                   1
1335                 1                   1
1336                 0                   0
1337                 0                   0

[1337 rows x 8 columns]
```

```
final_df.head()
```

```
        age  is_female       bmi  children  is_smoker  charges  \
0 -1.440418          1 -0.517949 -0.909234          1    16884
1 -1.511647          0  0.462463 -0.079442          0     1725
2 -0.799350          0  0.462463  1.580143          0     4449
3 -0.443201          0 -1.334960 -0.909234          0    21984
4 -0.514431          0 -0.354547 -0.909234          0     3866

   region_southeast  bmi_category_Obese
```

```
0                  0                    0
1                  1                    1
2                  1                    1
3                  0                    0
4                  0                    0
```

Machine Learning Starts

```python
from sklearn.model_selection import train_test_split

final_df.head(3)
```

```
        age  is_female       bmi  children  is_smoker  charges  \
0 -1.440418          1 -0.517949 -0.909234          1    16884
1 -1.511647          0  0.462463 -0.079442          0     1725
2 -0.799350          0  0.462463  1.580143          0     4449

   region_southeast  bmi_category_Obese
0                 0                   0
1                 1                   1
2                 1                   1
```

```python
X = final_df.drop('charges', axis=1)
y = final_df['charges']

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.20, random_state=42)
```

Linear Regression Model

```python
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, y_train)
```

```
LinearRegression()
```

```python
y_pred = model.predict(X_test)

from sklearn.metrics import r2_score

r2 = r2_score(y_test, y_pred)
r2

n = X_test.shape[0]
p = X_test.shape[1]
```

```
adjusted_r2 = 1 - ((1-r2) * (n-1) / (n-p-1))
adjusted_r2
```

```
0.7987962362937232
```